

VCU College of Engineering

CS 27-346: Automated Conversational Phone Agent for Credit Card Servicing

Preliminary Design and Proposal

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Capital One Sponsored Capstone Project

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Date

October 6th, 2026

Executive Summary

As a group of four Computer Science seniors in Virginia Commonwealth University, we will be collaborating with Capital One to develop an AI-powered agent designed to provide service for a consumer's credit card through phone calls without human intervention unless needed through certain circumstances. For this project one of our goals for this agent is for it to complete the following tasks including the authentication of the callers, resolving requests related to the account including filing disputes, giving out balance inquiries and sharing the status of payment. We are prompted to provide the agent natural speech responses as the person is responding to the phone agent as a way for it to have conversations with the caller as it completes these tasks through a sandboxed API.

For this semester, our team will learn the basics of AI agents and plan on some goals we would like to accomplish for this project. Some of the plans we will be focusing on will be setting up a speech/phone call model to interact with the agent, ensure the agent will be able to do the task we wanted it to complete, and ensure the agent will be interacting with the API and database in a secure and safe manner without causing catastrophic damage. We will also be planning on finding a cloud service that the agent can utilize and learn from to generate responses and execute actions through the mocked API, designed to be similar to an actual banking API for account management and verification.

The project is expected to require the research and experimentation of LLMs and automated account management being done outside of the need for human interventions in a secure and reliable manner. Capital One may benefit from this project with the potential to provide customer support and account management for consumers using their products when human agents are not available to provide these services in certain times. VCU and students of VCU can also learn how to create and handle AI agents with security, sustainability and performance in mind as it can be used as a tool with the potential to understand the complexity of certain problems and requests for the agent to search for ways to resolve them and provide the user the best solution possible while also respecting the information of the user interacting with the agent and the internet.

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Section A. Problem Statement

Credit card issuers serve millions of customers, and when those customers need help with their accounts, a lot of them still pick up the phone. Most of these calls are simple and are often calls that involve routine requests such as checking a balance, confirming a payment due date, making a payment, or reporting a lost card. Most companies use an interactive voice response (IVR) system, which is an automated phone menu where callers press buttons or say short commands. If the IVR can't resolve a request, the caller gets transferred to a human agent. Now this setup costs companies a lot of money. Gartner estimates that there are about 17 million contact center agents worldwide and that labor can account for up to 95% of contact center costs [3]. The model also frustrates customers. In a survey of more than 2,000 U.S. consumers, 61% said IVR menus create a poor customer experience, and 51% said that they had stopped doing business with a company because of its IVR [4]. The most common complaints were that the caller's reason for calling was not listed and that callers were forced to listen to options that did not apply to them [4]. Because of this, simple requests that should only take a few minutes often end up stuck in a queue waiting for a human agent. That drives up costs for the company and wait times for the customer.

To fix these problems, many companies are turning to conversational artificial intelligence (AI), which is software that understands natural speech or text and responds the way a person would. Instead of going through a menu, a caller can just explain what they need in their own words. Still, adoption has been slow. Gartner estimated that only about 1.6% of contact center agent interactions were automated with AI in 2022 and projected that this would grow to one in ten by 2026 [3]. A big reason is risk. In the financial industry, a system that gives wrong information or takes the wrong action can directly harm customers. Security experts also warn that AI systems allowed to take actions, like making a payment, can be manipulated into doing things they should not [7]. This shows a clear gap. Current systems are either safe but rigid, like IVRs, or flexible but hard to control, like many chatbots. Few can hold a natural conversation while making sure every account action is authorized, accurate, and compliant.

The Automated Conversational Phone Agent for Credit Card Servicing project is meant to fill this gap. The project is sponsored by Capital One, a U.S. bank based in McLean, Virginia. In May 2025, Capital One finished acquiring Discover Financial Services, which made it one of the largest credit card issuers in the U.S. by loan volume [5]. This project, built by computer science seniors at Virginia Commonwealth University, brings conversational AI to credit card servicing calls. The goal is a prototype agent that answers credit card servicing calls, verifies the caller's identity, understands natural speech, and handles common tasks like checking a balance, making a payment, or reporting a lost card. Every action will run against a mocked banking API with synthetic customer data and follow strict privacy and compliance rules. If a request is unsafe, unclear, or out of scope, the agent will transfer the caller to a human agent.

The main engineering challenge is balancing flexibility and control. The more freely an agent can understand and respond to natural speech, the harder it is to guarantee it only takes

safe, authorized actions. Each existing approach handles this trade-off differently. Touch-tone IVRs guide callers through fixed menus. They are predictable and easy to audit, but rigid, and callers often can't find their reason for calling in the options [4]. Speech-enabled IVRs let callers say short commands instead of pressing keys, but they still follow the same fixed menus. Intent-based bots, like those built with Amazon Lex, use automatic speech recognition (ASR) and natural language understanding (NLU) to interpret what a caller says [6]. Developers define intents, which are the tasks the bot can do, and slots, which are the details it needs to collect to finish a task [6]. This feels more natural and stays controlled, but it struggles with requests outside its predefined intents. Agents powered by large language models (LLMs) are the most flexible, but they bring new security risks. The Open Worldwide Application Security Project (OWASP) ranks prompt injection, where a user crafts input to manipulate the model, as the top risk for LLM applications [7]. OWASP also flags excessive agency as a major risk when an LLM can take actions like making a payment or changing account information [7]. Human agents give the best service and judgment, but they cost the most and are limited by staffing [3]. Each approach solves part of the problem, but none combines natural conversation with the assurance that every action is safe.

By pairing natural conversation with strict, rule-based control over every action, this project aims to show that a phone agent can be both helpful and trustworthy. The agent will use AI to understand what the caller wants, while a separate set of fixed rules decides whether each action is allowed. For example, the caller's identity must be verified before any account details are shared, and the caller must clearly confirm before a payment goes through. This gives Capital One a working model of how routine calls could be handled without a human agent, which could help cut the labor costs that make up most of contact center spending [3]. Cardholders get faster service at any hour, without menus or long hold times. The project also addresses the security risks of AI systems that can take actions by building safety and compliance into the system from the start [7]. Finally, it gives VCU students hands-on experience designing AI systems with security, privacy, and accountability in mind.

Figure 1. Traditional IVR Call flow

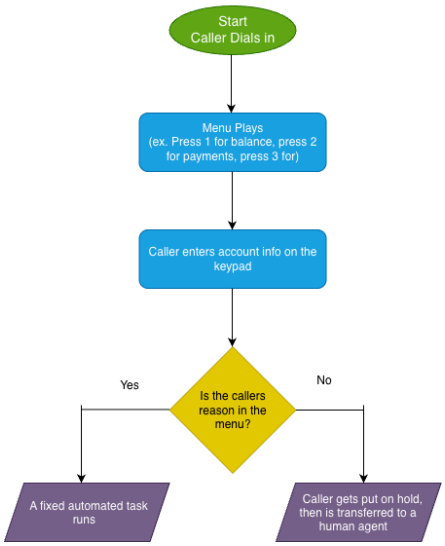
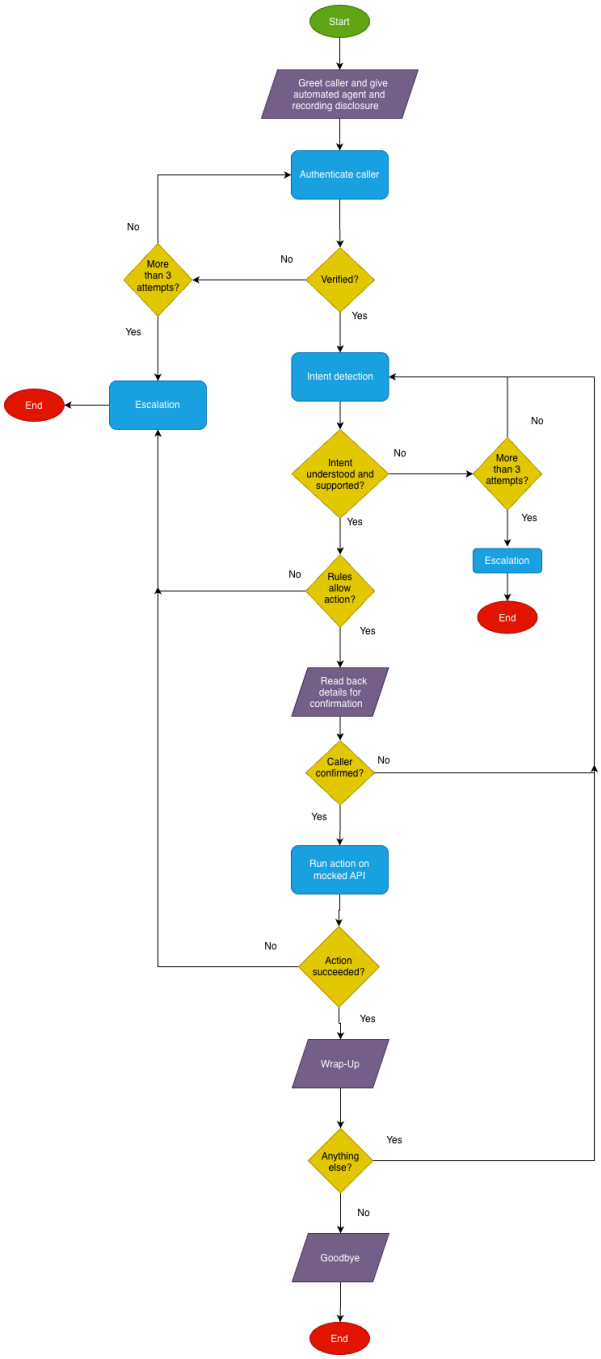


Figure 2. Conversational AI Phone Agent Call Flow



Section B. Engineering Design Requirements

The design requirements for this project are based on the project goals established with Capital One, research into conversational AI and autonomous AI agents, and the requirements of the VCU Computer Science capstone program. The system will investigate the use of an AI-driven telephone agent for credit-card servicing while emphasizing security, reliability, and appropriate human escalation.

B.1 Project Goals (i.e. Client Needs)

The primary goals of the project are to:

- Develop an AI-driven telephone agent using Amazon Connect.
- Demonstrate automated handling of common credit-card servicing requests.
- Provide secure customer authentication and authorization.
- Prevent the AI from performing unauthorized or unsafe actions.
- Protect sensitive customer and account information.
- Provide human escalation for requests that cannot be safely automated.
- Maintain an auditable record of servicing actions.
- Evaluate the AI agent against a traditional scripted IVR system.

All customer, account, and transaction information will be synthetic. The system will not connect to Capital One production systems or use real customer information.a

B.2 Design Objectives

The objectives of the project are:

1. Support at least 12 defined credit-card servicing scenarios.
2. Use Amazon Connect to receive and manage telephone interactions.
3. Authenticate callers before accessing protected account information.
4. Separate AI reasoning from the execution of account-changing actions.
5. Require additional confirmation and security controls for higher-risk transactions.
6. Escalate unsupported or high-risk requests to a human agent.
7. Protect sensitive information throughout the conversation and system logs.
8. Maintain an auditable record of actions performed by the system.
9. Support multi-step conversations while retaining relevant information.
10. Be evaluated using task success, containment, escalation accuracy, latency, and security metrics.
11. Achieve zero unauthorized actions during defined security testing.
12. Remain within the project's AWS cost limitations.

B.3 Design Specifications and Constraints

These constraints ensure that the framework remains scalable, open, and educationally extensible, while maintaining interoperability and realistic industry alignment.

Constraint	Type	Target/Description
Amazon Connect	Platform	System must use Amazon Connect for telephone communication.
AWS Services	Platform	System should primarily use AWS services available to the project.
Synthetic Data	Security	No real customer, account, payment, or transaction data may be used.
Project Budget	Cost	AWS costs should remain approximately 0–50$.
Project Schedule	Schedule	System must be completed within two academic semesters.
Telephone Access	Functional	System must support interaction through a real telephone number
AI Capabilities	Functional	System must support conversational AI for defined credit-card servicing tasks.

B.4 Codes and Standards

The project will be designed to follow the guidelines of the NIST SP 800 standards as a way to ensure the security and protection of personal information of the customer using the phone agent for services having to share their sensitive information to a chatbot.

Section C. Scope of Work

The goal of our project is to design and build a prototype phone agent that handles credit card servicing calls. A cardholder calls a real phone number, Amazon Connect answers, and the agent verifies who is calling, works out what the caller needs from natural speech, and completes the request itself. If a request is unsupported, the agent transfers the caller to a human. The work covers the call flow in Amazon Connect, the conversational agent, the fixed rules that approve or block each action the agent proposes, a roughly mocked banking API loaded with synthetic accounts, a record of every action taken, and the testing needed to show that the system meets the objectives in Section B.

The team will follow an agile, iterative approach. Work is planned in short cycles and reviewed every week in a meeting with the faculty advisor and the Capital One mentors. The fall semester centers on scope, design and documentation, together with early hands-on work in Amazon Connect. Most development and testing follows in the spring.

The following are outside the team's scope of work:

- Using real customer data or connecting to any Capital One production system.
- Automating the highest-risk requests, such as fraud claims, collections, hardship cases, and deceased or estate accounts.
- Supporting every servicing request a card issuer offers.

C.1 Deliverables

Technical deliverables

- Phone agent on Amazon Connect that answers a real phone number and handles the agreed servicing requests (balance checks, payments, disputes, etc.).
- Caller authentication before any account information is shared.
- Rule-based layer, separate from the AI model, that decides if an action is allowed.
- Transfer to a human agent for unsupported or high-risk requests.
- Mocked banking API with synthetic customer and account data.
- Audit log of every action the system takes.
- Compliance disclosures (such as call recording consent), added later in the project.
- Evaluation report based on the metrics in B.2.
- Simple dashboard for the demo.
- GitHub repository with source code and setup instructions.

Academic deliverables

- Team Contract (September 2026)
- Project Proposal (October 2026)
- Fall Design Poster and live poster session (November 2026)

- Preliminary Design Report (December 2026)
- Final Design Report (Spring 2027)
- Capstone Design EXPO poster and presentation (April 2027)

Potential risks and mitigation

- AWS Costs: Phone minutes and AI model usage are billed by use. We will keep test traffic small and set a billing alert.
- Shared AWS Account: Everything is hosted in one member's AWS account, so each member will get their own IAM user instead of sharing a login.
- Learning Curve: Amazon Connect is new to our team. We are building a basic call demo by October 22 to find problems early.
- Scope Creep: New requests or features need approval from the sponsor and faculty advisor before they are added.

C.2 Milestones

Spring dates are targets and may change as the project moves along.

Milestone	Description	Target Date
Proposal Submitted	Submit the project proposal	October 10, 2026
Sponsor Design Review	Present the design and a basic call demo to the Capital One mentors	October 22, 2026
Design Finalized	Agree on the request list, authentication steps and architecture	Early November 2026
Fall Design Poster	Present at the poster session	November 14, 2026
Preliminary Design Report	Submit report with the selected design	December 8, 2026
Prototype v1	Caller authentication and read-only requests on synthetic data	February 2027
Prototype v2	Payments and disputes with rule checks, human handoff and audit log	March 2027

Testing and Evaluation	Security testing, metrics, disclosures and demo dashboard	April 2027
Capstone EXPO	Final report, poster and live demo	April 24-25, 2027

C.3 Resources

For this project, we would need to rely mostly on software related resources.

Software and Digital Resources

- AWS - Amazon Connect Customer for hosting and developing the project
- Other AWS services - Amazon Lex, Bedrock, Lambda & DynamoDB
- GitHub for collaboration and version control of the project
- Figma

Human Resources & Guidance

- Four Senior Computer Science Students focusing on the Research, Development, Testing & Debugging.
- Faculty advisor providing technical oversight for the project and weekly progress reports.
- Capital One sponsor representatives offering feedback and guidance for the students working on the project.

Project Budget & Funding

- Estimated budget to be used: $50 - $250 to cover the use of AI tokens & service fees.

Section D. Concept Generation

A number of methods can be used to help generate design concepts from simple reflection and brainstorming, to working the problem backwards, using reverse thinking techniques, and looking to nature for inspiration (i.e. biomimicry). Existing solutions, or components of existing solutions, can be substituted, combined, adapted, modified, put to other uses, eliminated, or rearranged to meet new design objectives and specifications. A minimum of 3 overall design concepts is required for this section although more are welcome. Provide a brief description of how each design concept addresses the design problem. Discuss the potential pros and cons, including and potential risks of failure, of each of these concepts.

It is likely that each design concept may consist of several components. In this case, one or more of these components may offer a sub-problem that can be further explored, modified, or otherwise improved upon. These sub-problems may lead to the addition of several additional design concepts and may require the inclusion of a design concept chart or matrix to organize all ideas and potential solutions.

Provide any initial design sketches, drawings, 3D renderings, or conceptual models such as dataflow diagrams, process flows, etc. developed during the concept ideation phase. All hand drawings should be drawn to scale using basic engineering drafting tools (i.e. ruler, protractor, and compass). Geometric stencils can also be used to help produce quality hand drawings. Drawings should be presented in a profession manner, preferably done on engineering graph paper and using a high-quality scan. All sketches should be labeled to identify major components and different drawing views or projections if applicable. Basic dimensions should be provided to give a general sense of scale. Label each sketch or drawing with the name of the team member responsible for the sketch, the date it was drawn, and the drawing scale.

Section E. Concept Evaluation and Selection

Using a systematic decision-making process, evaluate each of the design concepts and choose the one that is most likely to succeed in meeting the design objectives and constraints. A Decision Matrix, or Pugh Matrix, helps to analyze alternatives, eliminate biases, and make rational decisions through thought and structure. First, work to develop a set of selection criteria for which to evaluate the previously generated design concepts. Selection criteria often include concepts of performance, cost, safety, reliability, risk, etc. Note that the selection criteria developed here will likely be more general than the project design objectives. As with the design objectives, conversations with the client help define appropriate selection criteria.

In many cases, the client may value the selection criteria differently, preferring that more emphasis be placed on some than others. In this case, weighting factors may be used to place more or less importance on the various criteria in the decision making process. Again, conversations with the client can be used to define criteria weighting factors. Often times, these conversations must be analyzed and interpreted by the team to determine which criteria are more important to the client and by how much. Feel free to discuss the assigned weighting factors with the client to see if they seem accurate.

Next, define an associated metric to represent each criteria. Metrics should be specific and quantifiable, providing numerical values that quantify the often vague concepts of the selection criteria. Metrics can be obtained, generated, or estimated through a number of methods including simple background research, preliminary design calculations, or basic analyses. Note that these metrics do not need to specifically align with the design specifications although there may be some commonality between the two. Provide a brief discussion of the rationale for selecting each of the assigned metrics.

Using the defined metrics, evaluated each design concept against all selection criteria by filling out a Decision Matrix. Design concepts can be compared by using simple rank scoring, raw scoring, or weighted scoring techniques and design concept with which to move forward can be selected. This type of process provides a meaningful, unbiased means for choosing a preliminary design concept prior to moving forward with more comprehensive, detailed analyses as provided in the design methodology section below. The results of this process should be discussed with the project client prior to moving forward with the selected design. Table 1 provides an example of a simple decision matrix.

Table 1. Example of a Decision Matrix.

	Design Concept A	Design Concept B	Design Concept C	Design Concept D
Criteria 1				
Criteria 2				
Criteria 3				
Criteria 4				
Criteria 5				
Total Score				

Note: Weights can be assigned to each criterion if desired.

Section F. Design Methodology

Provide a detailed explanation of the methods that will be used to help evaluate, improve, and evolve the design through the iterative engineering design process. Consider that ultimately, the final design must be verified and validated to ensure that it meets all of the previously developed and listed design objectives and specifications. Verification ensures that the design meets all specification, while validation confirms that the design functions as intended such to meet the client's needs. While it is common for initial design concepts to first be evaluated using simplified design criteria and metrics, the chosen design should be advanced, and later verified, using engineering calculations, computational models, experimental data, and/or testing procedures.

Use this section to describe any underlying physical principles and mathematical equations that govern the design. Provide details of any computer-aided modeling techniques used to evaluate the design including the software used, prescribed boundary conditions, and assumptions. Include a detailed description of any experimental testing methods including required testing equipment, test set-up layout, data acquisition and instrumentation, and testing procedures. If one or more prototypes is to be produced and tested, provide a detailed description of how each will be evaluated.

Note: The contents of this section are expected to vary from project to project. Subsections may be appropriate for providing details of analytical, computational, experimental, and/or testing methods. Some potential subsections that may be included in this section are provided. While critical design equations may be provided here, lengthy mathematical derivations may be included in an appendix. Validation procedures are critical and all projects should address such topic.

F.1 Computational Methods (e.g. FEA or CFD Modeling, example sub-section)

F.2 Experimental Methods (example subsection)

F.3 Architecture/High-level Design (example subsection)

F.5 Validation Procedure

Describe how the design team will validate that the final design meets the client's needs. This section should include a plan to meet with the client towards the end of the project to discuss final design details and demonstrate a prototype, experimental test, and/or simulation results. Provide a relative time frame for this validation to occur (e.g. "mid-March" or "early-April"). Include a brief discussion on how client feedback will be captured, such as a

formal survey, interview, or observation notes of the client using the prototype. It may also include plans to solicit feedback from other stakeholders and/or potential users.

Section G. Results and Design Details

Use this section to highlight the major results of the design methodology described above including important analytical, computational, experimental, modeling, assembly, and testing results. This section should be one of the most substantial sections of the report showcasing all of the hard work and effort that went into the completion of the final design and delivery of the project deliverables. Show how the identified problem was solved.

Highlight the prominent features of the final design through analysis results, modeling, drawings, renderings, circuit schematics, instrumentation diagrams, flow and piping diagrams, etc. to show that the design functions as intended and meets all design objectives and constraints. Overview designs such as dataflow diagrams, process flow, swim lane diagrams, as well as presentation-layer designs (e.g. storyboards for front-ends) should be included here. Detailed designs such as database designs, software designs, procedure flowcharts, or pseudocode should be included here. Support computational and experimental results with key plots and figures. All supporting figures should be clearly labeled and annotated to highlight the most important points of the figure (i.e. explicitly point out what the reader should focus on or understand about the image).

Note that while all results should be used to help inform design decisions, not all results may be necessary to include in the main body of the report. Extraneous supporting results (e.g. graphs, data, design renderings, drawings, etc.) that are not necessary for presenting the fundamental findings can be placed in one or more appendices. Detailed documentation of each program module can be provided as appendix.

G.1 Modeling Results (example subsection)

G.2 Experimental Results (example subsection)

G.3 Prototyping and Testing Results (example subsection)

G.4. Final Design Details/Specifications (example subsection)

Note that while the design constraints and specifications may have provided minimum or maximum values, or ranges or values, that the design needed to meet, the final design specifications should be listed here showing that the required design values were met. A list of final design details can also be included demonstrate fulfillment of the design objectives.

Note: Preliminary results should be included in the Preliminary Design Report to show the progress made of the selected design concept to-date. This section should be updated for the Final Design Report to include documentation of all of the work that was completed on the project throughout the entirety of the academic year.

Section H. Societal Impacts of Design

In addition to technical design considerations, contemporary engineers must consider the broader impacts that their design choices have on the world around them. These impacts include the consideration of public health, safety, and welfare as well as the potential societal, political/regulatory, economic, environmental, global, and ethical impacts of the design. As appropriate for the project design, discuss how each of these considerations influenced design choices in separate subsections. How will the design change the way people interact with each other? What are the political implications of the design? Does the technology have the potential to impact or shift markets? Does the design have any positive or negative effects on the environment? Don't forget to consider unintended consequences such as process or manufacturing byproducts. What impacts might the design have on global markets and trade? Are there any ethical questions related to the design?

While it is hard to forecast the various impacts of a technology, it is important to consider these potential impacts throughout the engineering design process. When considered during the early stages of the design phase, consideration of these impacts can help determine design objectives, constraints, and specifications and help drive design choices that may mitigate any potential negative impacts or unintended consequences.

Note: A minimum of 4 of these design considerations, including the consideration of public health, safety, and welfare, are required for the Preliminary Design Report while a section for all considerations must be included in the final design report.

H.1 Public Health, Safety, and Welfare

Provide a list of all design safety features and provide a brief description of each. Discuss the potential effects the design may have on public health, safety, and welfare. References to the codes and standards previous provided and the organizations that produced them may be summarized or referenced here.

H.2 Societal Impacts

H.3 Political/Regulatory Impacts

H.4. Economic Impacts

H.5 Environmental Impacts

H.6 Global Impacts

H.7. Ethical Considerations

Section I. Cost Analysis

Provide a simple cost analysis of the project that includes a list of all expenditures related to the project. If an experimental test set-up or prototype was developed, provide a Bill of Materials that includes part numbers, vendor names, unit costs, quantity, total costs, delivery times, dates received, etc. Do not forget to include all manufacturing costs incurred throughout the completion of the project. If the design is expected to become a commercial product, provide a production cost estimate including fixed capital, raw materials, manufacturing (including tooling and/or casting), and labor costs to produce and package the device. Note that this type of detailed cost analysis may be listed as a project deliverable.

Note: The Preliminary Design Report should include all costs incurred to date. It is expected that this section will be expanded and updated between the preliminary and final design reports.

Section J. Conclusions and Recommendations

Use this section to summarize the story of how the design team arrived at the final design. Focus on the evolution of the design through the use of the engineering design process including lessons learned, obstacles overcome, and triumphs of the final design. Revisit the primary project goals and objectives. Provide a brief summary of the final design details and features paramount to the function of the design in meeting these goals and objectives.

A discussion may be included to discuss how the design could be further advanced or improved in the future. If applicable, summarize any questions or curiosities that the final results/design of this effort bring to mind or leave unanswered. If this project might continue on as a future (continuation) senior design project, detail the major milestones that have been completed to date and include any suggested testing plans, relevant machine drawings, electrical schematics, developed computer code, etc. All relevant information should be included in this section such that future researchers could pick up the project and advance the work in as seamless a manner as possible. Documents such as drawings, schematics, and codes could be referenced here and included in one or more appendix. If digital files are critical for future work, they should be saved on a thumb drive, external hard drive, cloud, etc. and left in the hands of the project advisor and/or client.

Appendix 1: Project Timeline

Provide a Gantt chart of similarly composed visual timeline showing the start and end dates of all completed tasks and how they are grouped together, overlapped, and linked together. Include all senior design requirements including design reports and Expo materials (i.e. Abstract, Poster, and Presentation). All major milestones should be included in the timeline.

Appendix 2: Team Contract (i.e. Team Organization)

Copy and paste the content from the completed Team Contract here starting with Step 1 of the Team Contract and including all content following the ‘Contents’ list.

Appendix 3: [Insert Appendix Title]

Note that additional appendices may be added as needed. Appendices are used for supplementary material considered or used in the design process but not necessary for understanding the fundamental design or results. Lengthy mathematical derivations, ancillary results (e.g. data sets, plots), and detailed mechanical drawings are examples of items that might be placed in an appendix. Multiple appendices may be used to delineate topics and can be labeled using letters or numbers. Each appendix should start on a new page. Reference each appendix and the information it contains in the main text of the report where appropriate.

Note: Delete this page if no additional appendices are included.

References

Provide a numbered list of all references in order of appearance using APA citation format. The reference page should begin on a new page as shown here.

[1] VCU Writing Center. (2021, September 8). *APA Citation: A guide to formatting in APA style.* Retrieved September 2, 2024. https://writing.vcu.edu/student-resources/apa-citations/

[2] Teach Engineering. *Engineering Design Process*. TeachEngineering.org. Retreived September 2, 2024. https://www.teachengineering.org/populartopics/designprocess

[3] Gartner. (2022, August 31). *Gartner predicts conversational AI will reduce contact center agent labor costs by $80 billion in 2026*. https://www.gartner.com/en/newsroom/press-releases/2022-08-31-gartner-predicts-conversational-ai-will-reduce-contac

[4] Vonage. (2019, September 26). *Vonage research reveals IVR horror costs businesses $262 per customer each year.* https://www.vonage.com/about-us/newsroom/press-releases/Vonage-research-reveals-IVR-horror-costs-businesses-262-per-customer-each-year-09-26-2019/444cff58-7541-43bd-b08b-41d47951117d

[5] PYMNTS. (2025). *Capital One finalizes $35 billion Discover purchase.* https://www.pymnts.com/acquisitions/2025/capital-one-finalizes-35-billion-discover-purchase/

[6] Amazon Web Services. (n.d.). *Amazon Lex V2 core concepts.* In Amazon Lex V2 developer guide. https://docs.aws.amazon.com/lexv2/latest/dg/how-it-works.html

[7] OWASP Foundation. (2024). *OWASP top 10 for LLM applications 2025*. https://genai.owasp.org/llm-top-10/